

Brainstorming & Idea Prioritization Template

Date	04-07-2026
Team ID	
Project Name	Opticrop
Maximum Marks	3 Marks

Step 1: Brainstorm and Idea Listing

Each team member lists out as many ideas as possible without judging them at this stage.

S.No	Team Member	Idea / Suggestion	Category	Group No.
1	D.Goutham			

2. Project Overview

Purpose

OptiCrop is a machine learning-based agricultural recommendation system designed to optimize crop production using environmental and soil parameters. It integrates ML algorithms with a Flask web application to provide intelligent crop recommendations based on key environmental factors, helping farmers, agricultural researchers, agribusiness companies, and policymakers make data-driven decisions that improve productivity, sustainability, and resource utilization.

Features

- ? Smart crop recommendation based on soil nutrients (N, P, K) and environmental parameters
- ? Crop suitability and environmental assessment for a given set of conditions
- ? Support for agricultural research and policy planning through data-driven pattern analysis
- ? Comparison of multiple ML algorithms (KNN, Logistic Regression, Decision Tree, Random Forest, K-Means) to select the best-performing model
- ? Interactive, responsive web interface for entering parameters and viewing instant predictions
- ? Trained model persistence using Pickle (.pkl) for fast reuse during prediction